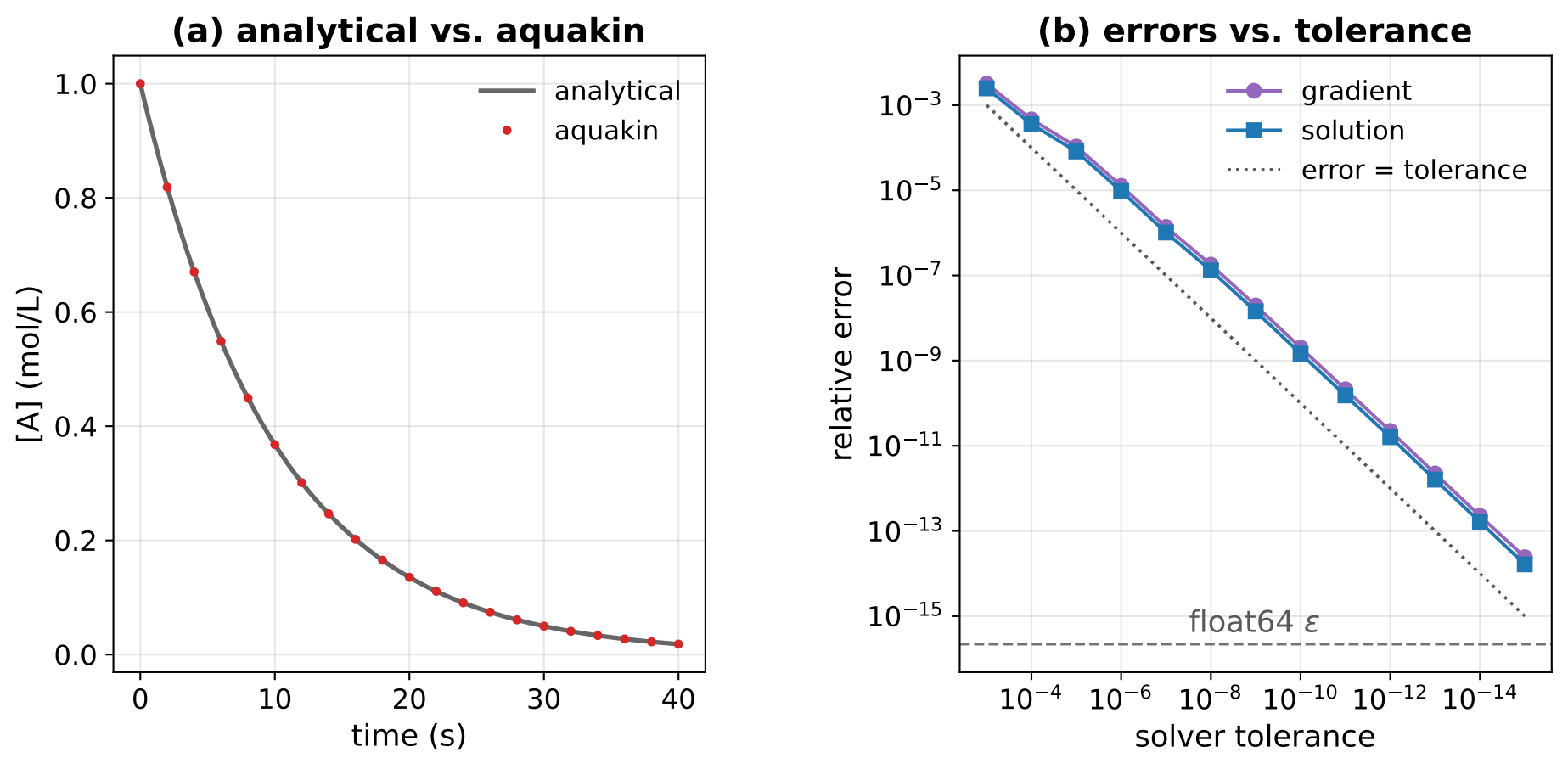
(a) analytical vs. aquakin
analytical
aquakin
[A] (mol/L)
time (s)
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
(b) errors vs. tolerance
gradient
solution
error = tolerance
relative error
solver tolerance
float64 ε
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}